

University of Birmingham Observatory
Request for Observing Time
Cycle 13: October 27 – November 28, 2025

Title: How Can Observations of Globular Clusters Help Estimate the Age of the Universe?
Group ID: 97547
Abstract: Globular clusters are tightly bound dense spherical clusters that formed in the very early days of the universe. Due to the lack of dust and nebulae being able to get in, new stars are very rarely formed meaning all the stars in the cluster are roughly the same age. This means by measuring the cluster and resolving the stars in it, the age of stars just turning off the main sequence can be determined by comparing them to stellar evolution models which roughly tell us the age of the cluster. This project aims to do this for cluster NGC 6341, as it is one of the only clusters visible at this time of year and bright enough to resolve with our telescope. By resolving the stars in 2 different filters, we can plot a Colour Magnitude diagram (CMD) and use that to find the turn off stars.
Category: Regular
Time Requested: 10 hours
Moon Phase: Grey

Summary of Targets and Observations
 (continue on last page if more space is needed)

Target name	RA (J2000)	Dec (J2000)	Filter	Time (hrs)	Comment
NGC-6341	17:17:07.39	+43:08:09.4	V	2.5	Start of the night, early in the month of November.
NGC-6341	17:17:07.39	+43:08:09.4	B	2.5	Start of the night, early in the month of November.

Scientific Justification: (Maximum 1 page)

In the early stages of the universe, individual dense clouds of gas (nebula) collapsed to form anywhere from thousands to millions of stars (Kruijssen 2025). These vast collections of stars, otherwise known as globular clusters, are typically gravitationally bound in very dense spherical shapes, and are found in the outer regions of galaxies such as the Milky Way; roughly 170 globular clusters are present in our galaxy (Massari 2025). After their formation, globular clusters lack the required collections of dust and gas to be able to form new stars over time, therefore, when observing globular clusters, it is correct to assume all stars inside the cluster formed at approximately the same time (ESA/Hubble n.d.). Observations of globular clusters require higher precision when identifying individual star components, particularly when looking at dimmer stars. This is due to the very high number density of stars present in globular clusters in a relatively small observation region. When compared to other types of clusters, such as open clusters, this clearly outlines why observing globular clusters is so necessary from an astronomy standpoint, mainly due to every star being essentially the same age. This research project will use this key fact as a backbone to help observe and analyse data from the globular cluster NGC-6341.

There are several areas of interest that can be explored from observations regarding globular clusters (Gratton et al. 2019). This research project will attempt to explore one of these areas in full detail, that being, calculations to obtain an estimate of the age of the Universe (Valcin et al. 2025). Other key research areas that can be discussed include investigations into stellar evolution as well as observing chemical compositions of globular clusters.

Each of these topics are important areas of research for both the specific field of globular clusters as well as astronomy in general. Certain results, particularly an estimate on the age of the Universe, can be correlated across to other areas of astronomy. It's clear that comparing an age obtained using globular cluster observations to one obtained through uses of Hubble's law and the Hubble constant (MacCallum 2015) would help to affirm this method as a generally accurate one. Hence allowing for a general cosmic timeline to be plotted which can be used throughout the whole field of astronomy when making observations regarding formations of stellar objects throughout the Universe.

Analysing globular clusters allows for stellar evolution to be studied. This is in essence a description of the life cycle of a star, showing how it can undergo changes over time (Aguirre 2017). Since the stars in globular clusters are assumed to have formed at the same time, age can be treated as a constant. Hence, the actual physical state of the star can be analysed and related to simply the star's mass (Catelan et al. 2009). This is seen on a Hertzsprung-Russell diagram (HRD), where the massive, brighter stars live for much shorter periods when compared to smaller, dimmer stars. This is one of the most important areas of astronomy as it provides evidence into the entire lifetimes of stars and reinforces research into the end states of stars which are dependent on their mass, those final states being: neutron stars, black holes, white dwarfs.

Looking at chemical compositions of globular clusters via spectroscopy allows for metallicity of globular clusters to be determined. Metallicity is a measurement of the number of elements heavier than Hydrogen (H) and Helium (He) in a stellar body (Latour et al. 2025). This is relevant to astronomy in general as it helps prove that in early stages of the Universe metallicity was low, mainly since many stars in comparison to the present day had not formed yet. As a result, these heavier elements were not as abundant.

As previously mentioned, the focus of this research project is to initially observe and reduce data from cluster NGC-6341. This data reduction would provide the magnitudes and colour indices of stars in the cluster. Consequently, this would allow for a Colour-Magnitude diagram (CMD) to be plotted. As colour is correlated to temperature, and magnitude to luminosity, this CMD is fundamentally a HRD for the globular cluster being observed. HRDs help portray a star's life depending on its temperature and luminosity. By looking at the main sequence turn off point (MSTO) on the produced HRD and comparing it to a stellar evolution model, an age for the cluster can be calculated (Spergel et al. 1997). Hence, assuming clusters formed in the very early stages of the universe, we can also estimate an age for the Universe.

Technical Justification: (Maximum 1 page)

Feasibility calculation:

The observation should result in conclusions about the age of a given cluster therefore there is less purpose in observations of more than one cluster looking for general conclusions. It is for this reason a two-filter analysis (visual and blue bands) of a single cluster was decided so the time of observation can be maximised. Because we are plotting a CMD, we need to be able to observe as many high magnitude stars as possible ($m=19$) resulting in a signal to noise ratio that was limited by a reasonable time of observation. With the help of some code to speed up our time to calculate, a signal to noise ratio of five for an apparent magnitude of 19 will take 2.5 hours under a grey moon phase. This was quickly seen as the best scenario due to how quickly the time of observation increases for higher magnitude or a larger signal to noise ratio. Anything more ambitious and the time becomes unrealistic. This time of observation calculation includes the presence of light cloud cover causing a 20% extinction to the flux received from the cluster.

The optimum time to observe our target is the beginning of November due to it being high in the sky for longer into the night meaning observations can be made looking through less airmass. This changes by the end of November as the cluster does not get as high in the sky while also dipping to angles with a high air mass earlier giving less time for good observations.

Analysis plan:

After we have reduced the data using our calibration plans, we now have the images we took saved in a .fits file. Turning this into a catalogue of stars would be incredibly time consuming to do by hand, so we must automate. We will use a program like SExtractor (Barbary 2016) to automatically go through the image and turn any bright points into a different star. Now that we have the catalogues, we can set up CMD plots by plotting the magnitude of each individual star in the visual band against the difference between the magnitude in the visual and the blue band. This represents magnitude and colour, and since colour depends on temperature, we now have an equivalent to a HRD. Finally, we can determine the point at which the stars start to leave the main sequence on the HRD and comparing them to a stellar model based on their size and temperature we can work out how old those stars are which will give us the age of the cluster.

Calibration plan:

Before the analysis of our data, we will take precise precautions to reduce the data and noise, by using several calibration frames to correct the readings. The initial step would be to correct for the inherent offset of the CCD, the bias voltage. To amend this, we will take zero-second exposures whilst the shutter is closed, and obtain an average over multiple readings, which will be removed from all frames. Secondly, we will attempt to correct for the dark current, this will be done by taking an exposure of equivalent time (2.5 hours for each filter) to the scientific frame with the shutter closed. This dark frame, which has already been reduced by bias, will be removed from the scientific frame to reduce noise from the detector rather than the observable object. Ultimately, we will correct for the variation of sensitivity within each pixel, imperfections in the optics and dust by taking a flat frame. For the flat frame, we will be taking a short observation of the sky during the twilight sky where the source is uniformly illuminated. This will be divided into the reduced scientific frame, where the result will be free of any systematic errors caused by the instruments.

Due to NGC-6341's position in the night sky (RA ~ 17 hours), the observation time would have to be taken quite early in the night, starting observations around 18:00 and ending by 21:00. As a result, our calibration frames, apart from the flat frame, would be taken after the observations. This would not cause any issues as both the bias and dark frame are taken with the shutter closed, so are independent of the night sky and observations.

Figures and tables (optional): (Maximum 1 page)

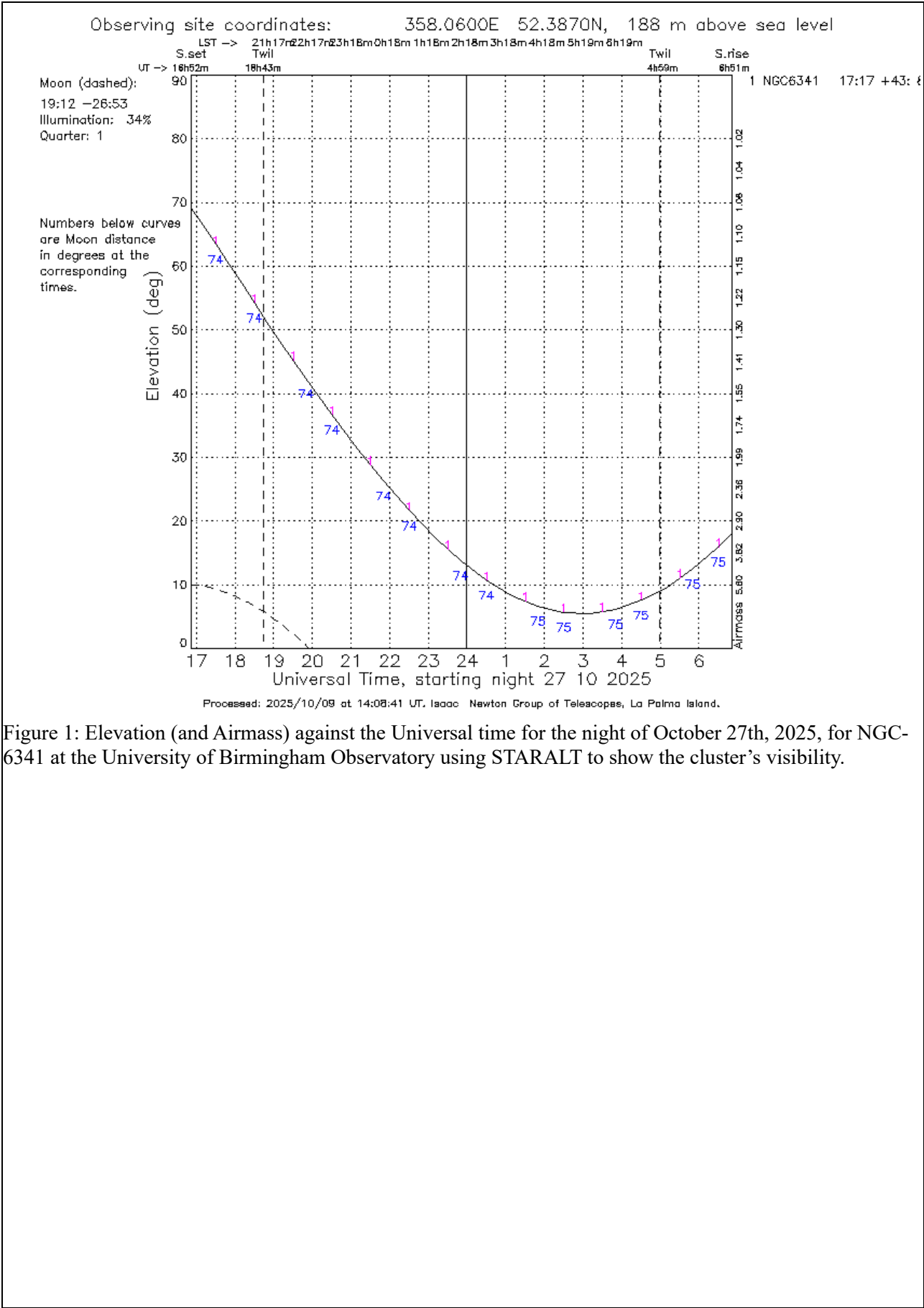


Figure 1: Elevation (and Airmass) against the Universal time for the night of October 27th, 2025, for NGC-6341 at the University of Birmingham Observatory using STARALT to show the cluster’s visibility.